PHYS 230 - Electricity and Magnetism
Lecture 26

Instructor: Saeed Rastgoo

UNIVERSITY OF ALBERTA

Outline

6. Direct-Current Circuits
   - Kirchhoff's Rules
   - $R - C$ Circuits

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

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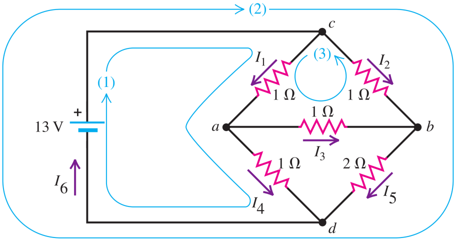
Exercise 1 https://epoll.srv.ualberta.ca – Code: MBD

Figure shows a "bridge" circuit. Find the current in each resistor and the equivalent resistance of the network of five resistors.



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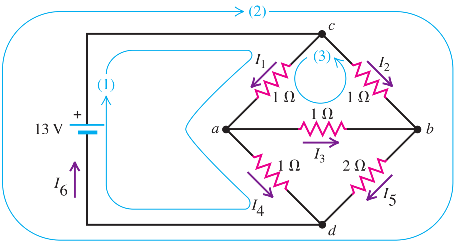

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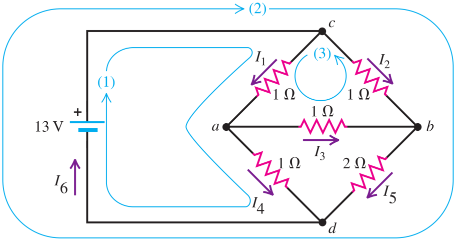

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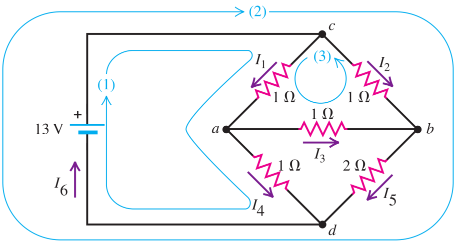

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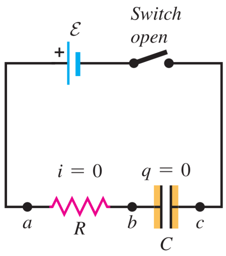
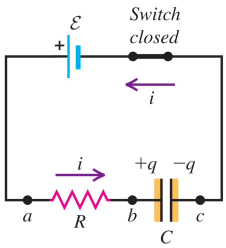
$R - C$ Circuits: Charging Capacitors

A circuit with a battery, a resistance and a capacitor is called an $R - C$ circuit

(a) Capacitor initially uncharged



(b) Charging the capacitor



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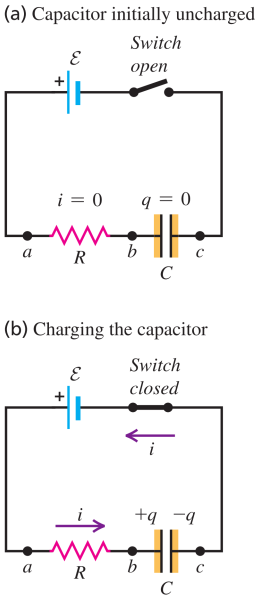

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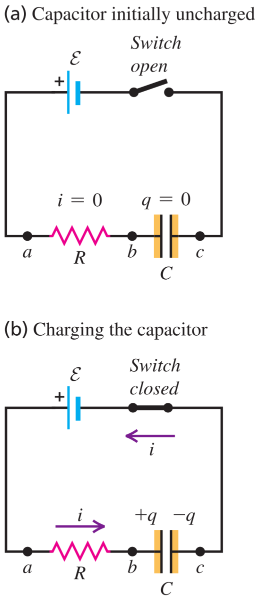
Initially the switch is open. At $t = 0$, we close the switch, current around the loop begins charging the capacitor



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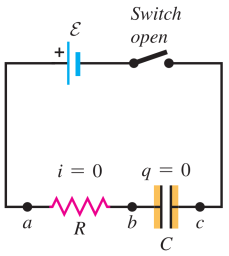

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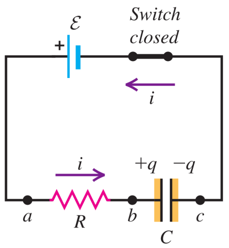

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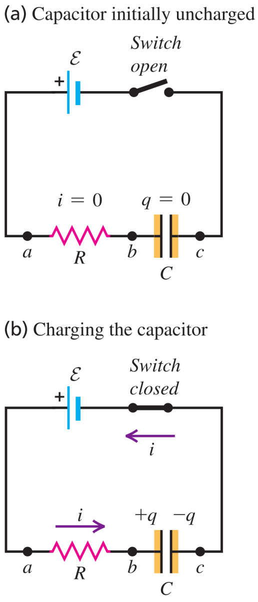
$$v_a - I_0 R - \frac{q_0}{v_{cb}} + \mathcal{E} = v_a$$



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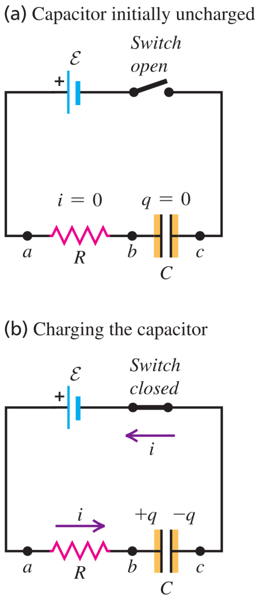

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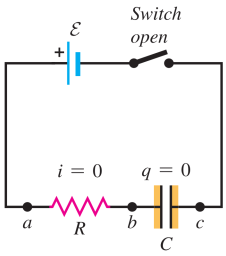
$$V_a - I_0 R - \frac{q_0}{V_{cb}} + \mathcal{E} = V_a \Rightarrow I_0 = \frac{\mathcal{E}}{R}$$

As the capacitor charges, V_{cb} increases, and since $V_{ba} + V_{cb} = \mathcal{E}$, V_{ba} decreases due to a decrease in current.

(a) Capacitor initially uncharged



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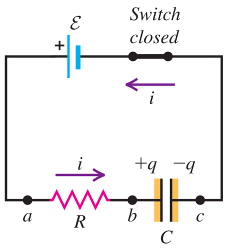

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As the capacitor charges, v_{cb} increases, and since $v_{ba} + v_{cb} = \mathcal{E}$, V_{ba} decreases due to a decrease in current.

After a long time the capacitor is fully charged, $I \to 0$, and $v_{ba} \to 0$. Then $v_{cb} = \mathcal{E}$.

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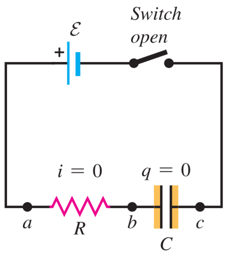

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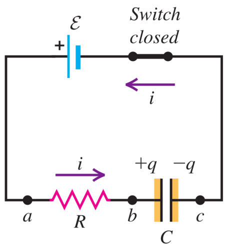

$R - C$ Circuits: Charging Capacitors

At $t > 0$, but before the capacitor is fully charged, let q represent the charge on the capacitor and i the current in the circuit

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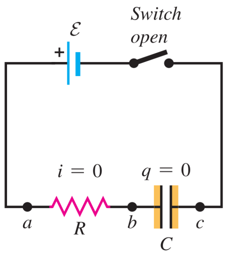

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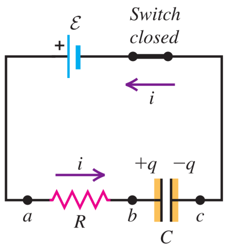

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At $t > 0$, but before the capacitor is fully charged, let q represent the charge on the capacitor and i the current in the circuit

The instantaneous potential differences v_{ba} and v_{cb} are

$$v_{ba} = iR, \qquad v_{cb} = \frac{q}{C}$$

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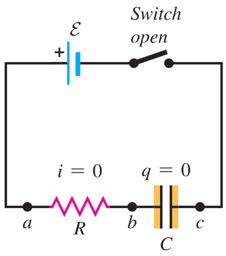

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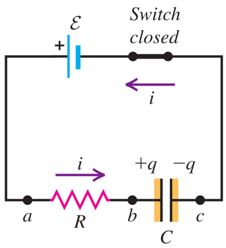

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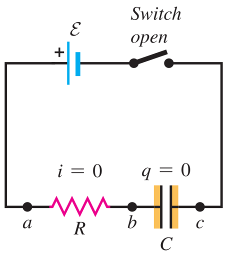

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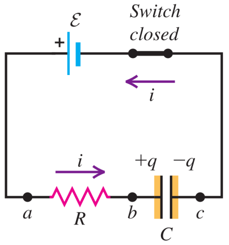

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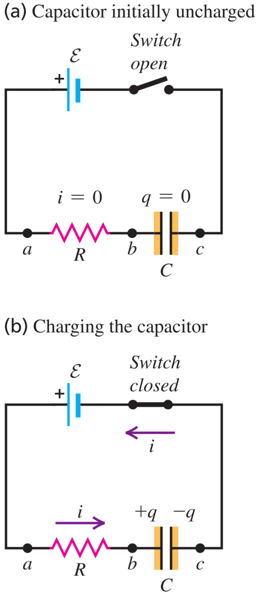

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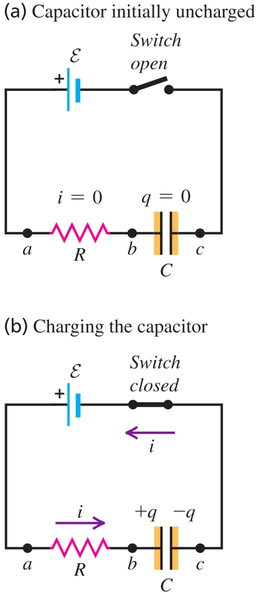

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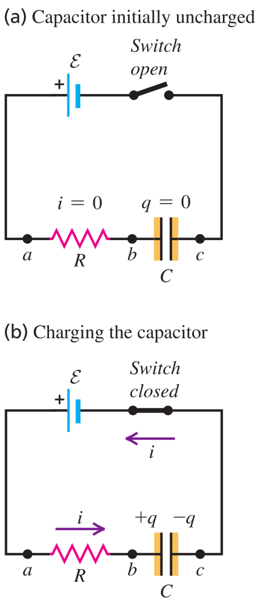

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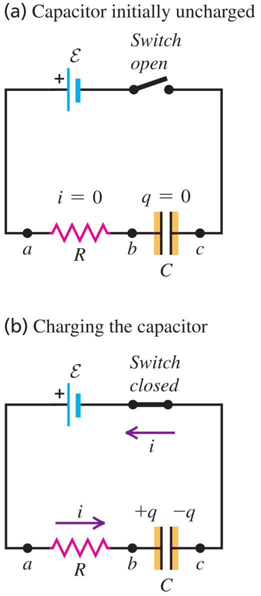
Rearranging this equation we get

$$\frac{dq}{q - C\mathcal{E}} = -\frac{1}{RC}dt$$



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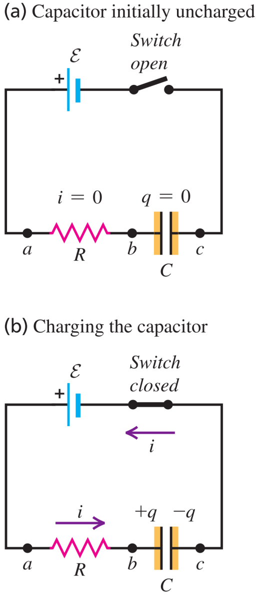
Rearranging this equation we get

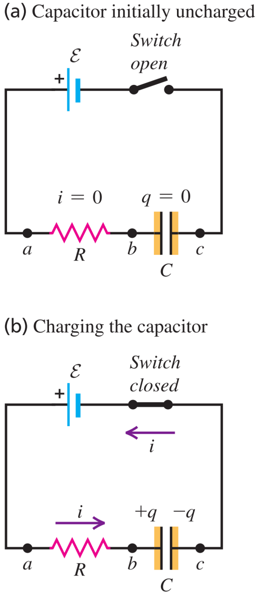
$$\frac{dq}{q - C\mathcal{E}} = -\frac{1}{RC}dt \Rightarrow \int_0^q \frac{dq}{q - C\mathcal{E}} = -\frac{1}{RC}\int_0^t dt$$



$R - C$ Circuits: Charging Capacitors

We can solve this integral using substitution $u = q - C\mathcal{E} \Rightarrow du = dq$, where the integral limits change to $-C\mathcal{E}$ to $q - C\mathcal{E}$

$$\left[\ln(u)\right]_{-C\mathcal{E}}^{q-C\mathcal{E}} = -\frac{t}{RC}$$



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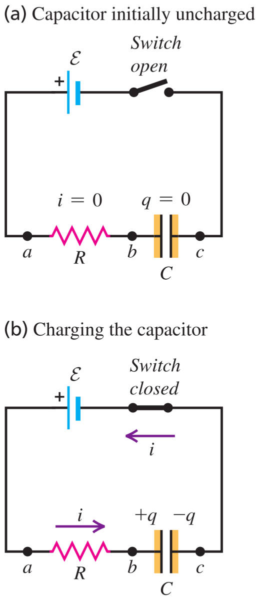

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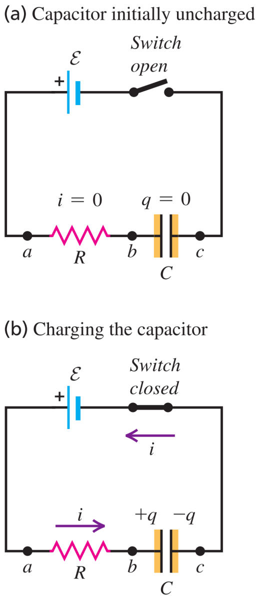

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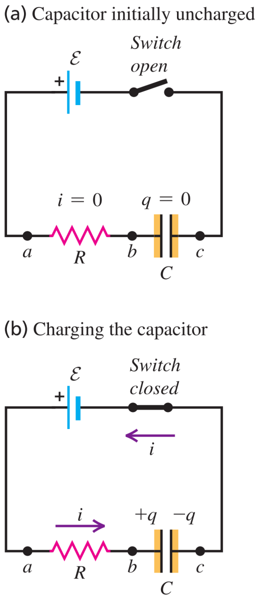
And exponentiating both sides we get

$$\boxed{q = C\mathcal{E}\left(1 - e^{-\frac{t}{RC}}\right)}$$



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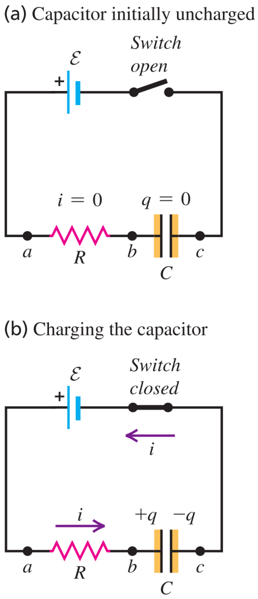
When a long time passes $t \to \infty$ we get the full charge

$$Q_f = C\mathcal{E}\left(1 - e^{-\frac{\infty}{RC}}\right) \Rightarrow Q_f = C\mathcal{E}$$



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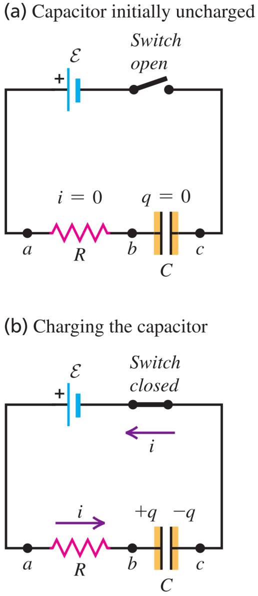
When a long time passes $t \to \infty$ we get the full charge

$$Q_f = C\mathcal{E}\left(1 - e^{-\frac{\infty}{RC}}\right) \Rightarrow Q_f = C\mathcal{E}$$

So we can write the charging equation

$$\boxed{q = Q_f\left(1 - e^{-\frac{t}{RC}}\right)}$$



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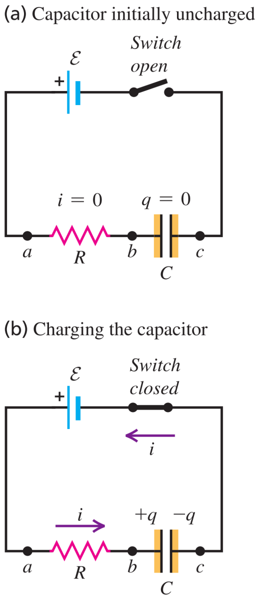
Taking derivative of both sides w.r.t. t

$$\frac{dq}{dt} = C\mathcal{E}\left(-\frac{d}{dt}e^{-\frac{t}{RC}}\right)$$



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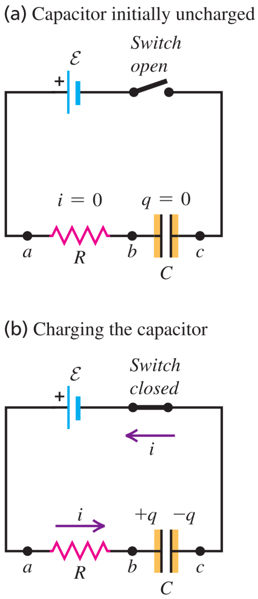
$$\frac{dq}{dt} = C\mathcal{E}\left(-\frac{d}{dt}e^{-\frac{t}{RC}}\right) \Rightarrow i = \frac{C\mathcal{E}}{RC}e^{-\frac{t}{RC}}$$



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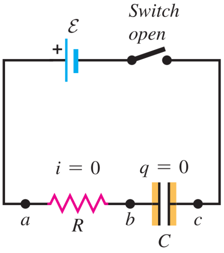

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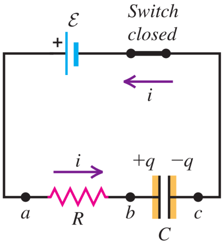

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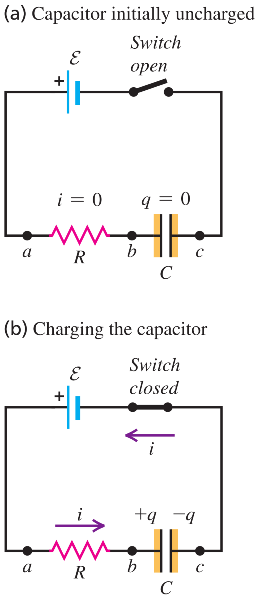
Taking derivative of both sides w.r.t. t

$$\frac{dq}{dt} = C\mathcal{E}\left(-\frac{d}{dt}e^{-\frac{t}{RC}}\right) \Rightarrow i = \frac{C\mathcal{E}}{RC}e^{-\frac{t}{RC}} \Rightarrow \boxed{i = \frac{\mathcal{E}}{R}e^{-\frac{t}{RC}}} \Rightarrow \boxed{i = I_0 e^{-\frac{t}{RC}}}$$



$R - C$ Circuits: Charging Capacitors

Taking derivative of both sides w.r.t. t

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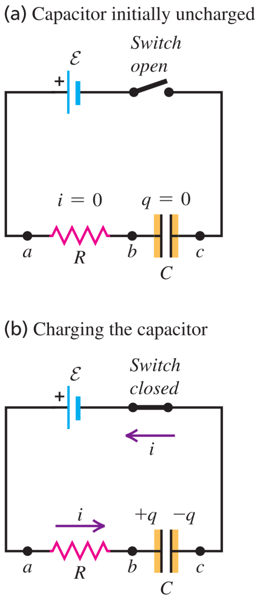
When $t = RC$, we get

$$q = C\mathcal{E}\left(1 - e^{-1}\right) = C\mathcal{E}\left(1 - \frac{1}{e}\right) = 0.632C\mathcal{E} = 0.632Q_f$$



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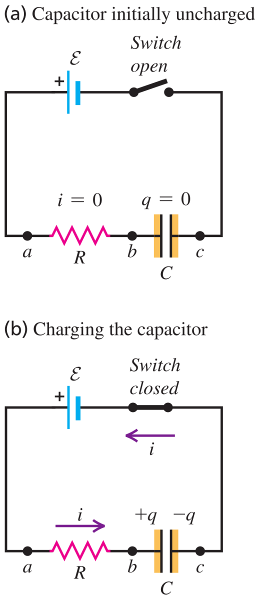
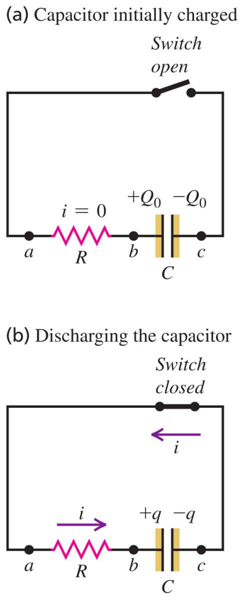
The **time constant** or the **relaxation time** τ of the circuit is defined as the time after the charge/current in the $R - C$ circuit reaches 0.632 its final value

$$\boxed{\tau = RC}$$



$R - C$ Circuits: Discharging Capacitors

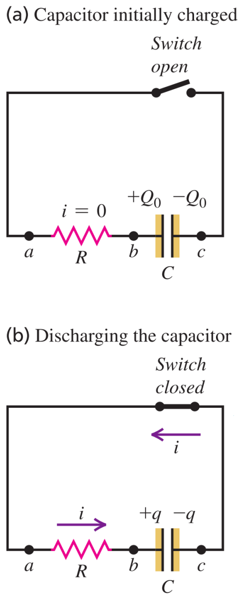
Now we remove the battery, replace it with an open switch, reset our time, and close the switch; at this time $q = Q_0$



$R - C$ Circuits: Discharging Capacitors

Now we remove the battery, replace it with an open switch, reset our time, and close the switch; at this time $q = Q_0$

The capacitor then discharges through the resistor, and its charge eventually decreases to zero.



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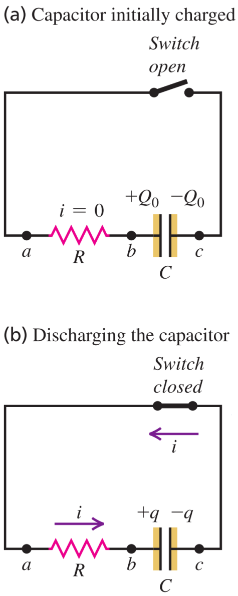
During discharge, using Kirchhoff's loop rule we get

$$V_a - iR - \frac{q}{C} = V_a$$



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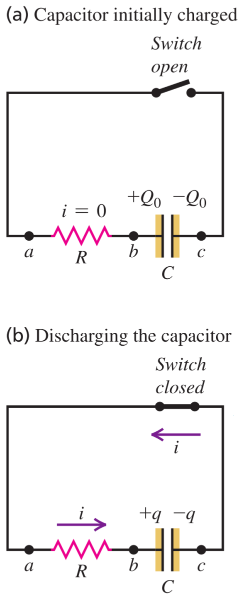
During discharge, using Kirchhoff's loop rule we get

$$V_a - iR - \frac{q}{C} = V_a \Rightarrow i = -\frac{1}{RC}q$$



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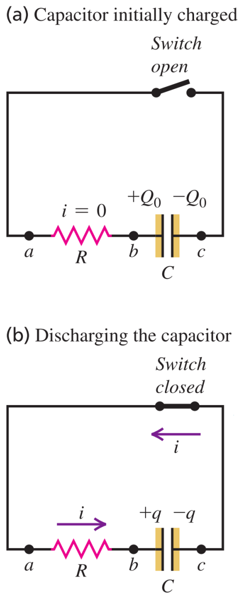
$$V_a - iR - \frac{q}{C} = V_a \Rightarrow i = -\frac{1}{RC}q \Rightarrow \frac{dq}{dt} = -\frac{1}{RC}q$$



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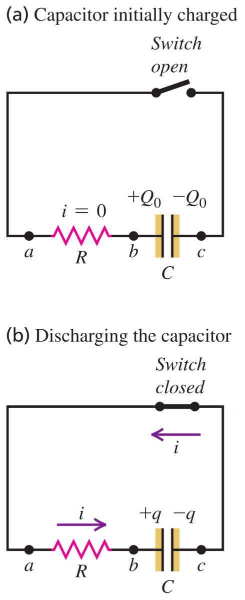
Rearranging this equation we get

$$\frac{dq}{q} = -\frac{1}{RC}dt$$



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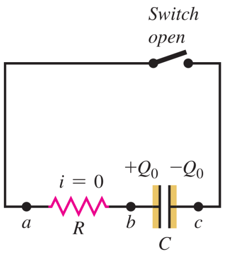
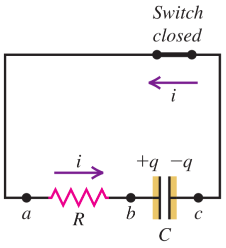
Rearranging this equation we get

$$\frac{dq}{q} = -\frac{1}{RC}dt \Rightarrow \int_{Q_0}^{q} \frac{dq}{q} = -\frac{1}{RC}\int_0^t dt$$

(a) Capacitor initially charged



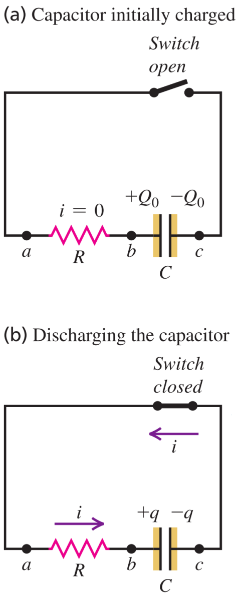
(b) Discharging the capacitor



$R - C$ Circuits: Discharging Capacitors

We can solve this integral as

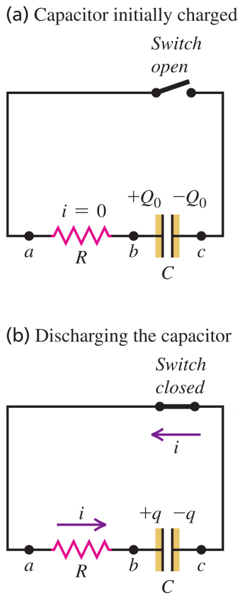
$$[\ln(q)]_{Q_0}^{q} = -\frac{t}{RC}$$



$R - C$ Circuits: Discharging Capacitors

We can solve this integral as

$$\left[\ln(q)\right]_{Q_0}^{q} = -\frac{t}{RC} \Rightarrow \ln(q) - \ln(Q_0) = -\frac{t}{RC}$$



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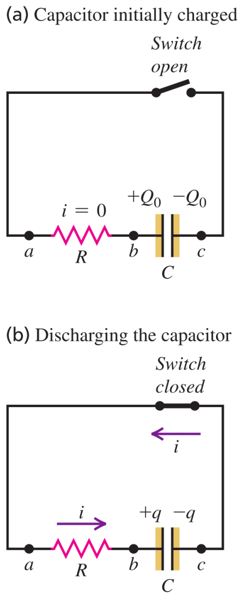
We can solve this integral as

$$[\ln(q)]_{Q_0}^{q} = -\frac{t}{RC} \Rightarrow \ln(q) - \ln(Q_0) = -\frac{t}{RC} \Rightarrow \ln\left(\frac{q}{Q_0}\right) = -\frac{t}{RC}$$



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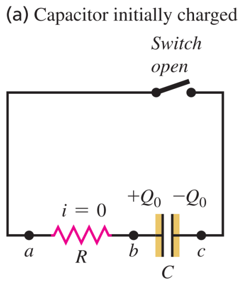
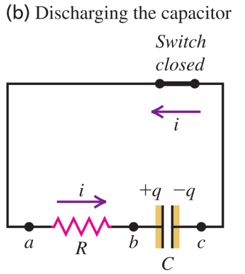
And exponentiating both sides we get

$$\boxed{q = Q_0 e^{-\frac{t}{RC}}}$$





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When a long time passes $t \to \infty$ we get the full charge

$$Q_f = 0$$

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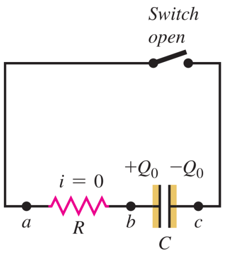

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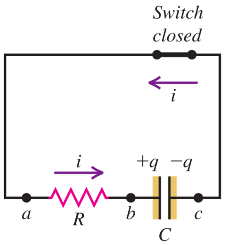

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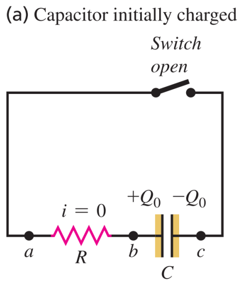
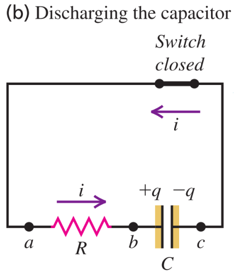
Taking the time derivative of both sides yields the current equation

$$\frac{dq}{dt} = Q_0 \frac{d}{dt} e^{-\frac{t}{RC}}$$





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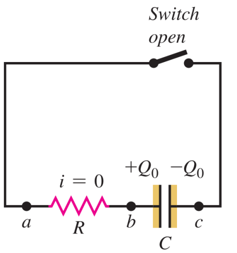

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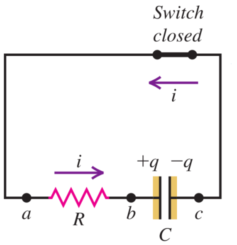

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We can solve this integral as

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